QSS 20 Final Project Milestone 2

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GitHub Code Repository: https://github.com/ConnorStephen/QSS-20-Final-Project-Milestone-1---Gavin-and-Connor

- At this stage, you should have (or be close to having) a working data processing pipeline, some initial results, or at least data exploration. The checkpoint is meant to validate your approach and ensure some progress has been made.
- Push some of your work-in-progress code to your GitHub repository, and add a `README.md` file to your repository explaining what is in the repository. Here is an example GitHub repo you can model the organization of your repo after: [Link].
- Comment out the blue bullet points and the Typesetting Examples in Section **??** before submission.
- Submit this PDF on Gradescope by **March 3, 2026**.

1 Introduction

The goal of our research is to find out whether court surface has driven specialization in tennis, and if modern tennis matches have become more serve dominated and court adapted over time. In order to accomplish this we will be tracking the stats and performances on different courts from top players of different time periods. We will compare their serve metrics and playstyles to see if modern players exhibit more adaptable and serve heavy numbers than their past counterparts.

2 Related Work

There have been many articles that take into account the types of court, serving numbers, and their impact. Works like “Nash at Wimbledon” (Gauriot and Page, 2016), which models serve direction as a mixed-strategy equilibrium. Other work has examined surface effects on match outcomes and point structures, showing that clay courts historically reduce serve effectiveness relative to grass. ”Physical Demands of Tennis Across the Different Court Surfaces, Performance Levels and Sexes: A Systematic Review with Meta-analysis” by Pluim et al compares performances of different genders on each of the different court types, based in effort and distance traveled versus rest. Jeff Sackmann’s Tennis Abstract database has also documented longitudinal trends in serve speeds, ace rates, and rally length. These studies do incorporate aspects of our research but none of them go as far in depth as our research. Our work is to compare court impact and playstyle over a much broader time frame than most of these other works. Our project will be integrating surface specialization, longitudi-

nal serve trends, and unsupervised clustering of play styles into a unified framework; with the range of our data going back as far as the 1980's up to modern day matches played in 2025.

3 Data

Our project uses Jeff Sackmann / Tennis Abstract "MatchChart" men's charting data. We currently have three point-by-point CSV files covering all charted matches from 1960–2009, the 2010s, and the 2020s, plus a match-level metadata table (`charting-m-matches.csv`) that includes match date (year), surface (Hard/Clay/Grass), tournament, round, and player names. We merge point-level data to match metadata on `match_id`, producing a unified dataset with point outcomes and surface/year labels. In preprocessing, we construct a single `point_string` for each point by using the `2nd` column when present (indicating a missed first serve and the rally recorded on the second serve) and otherwise using the `1st` column. We then parse `point_string` into features needed for analysis, including serve direction (`serve_dir`), indicators for aces and unreturnables (`is_ace`, `is_unreturned`), a serve-and-volley flag (`is_sv`), rally length (`rally_shots`), a point-ending label (`point_end`: winner / unforced error / forced error / serve_only / unknown), and an approach indicator (`has_approach`). To make analysis feasible and reproducible, we also build an aggregated dataset (`outputs/player_surface_year.csv`) with one row per (player, year, surface, role), where role is server or returner; this file is the primary input for our trend, specialization, and (later) clustering analyses. In terms of scale and composition, our merged parsed dataset contains roughly 1.22 million points across about 7.1 thousand matches spanning 1960–2025. Hard-court points make up the largest share, followed by Clay and Grass, and the distribution of `point_end` is dominated by forced errors, unforced errors, and winners, with a substantial share classified as serve-only points and a small residual classified as unknown due to edge-case strings.

4 Methodology

Our methodology follows the staged plan from Milestone 1: we first create interpretable point-level features from the MatchChart strings, then aggregate those into player-year-surface summaries, and finally use those summaries to study surface specialization and serve dominance over time (with clustering as the next milestone). From the parsed point-level features, we compute service-dominance metrics on server-side points such as serve-only rate (the share of service points ending immediately at the serve, as captured by our `serve_only` label), ace rate (share of service points that are aces), and average rally shots (mean rally length proxy). We compute surface-by-year trend lines using weighted averages, where weights equal the number of charted points in each player-year-surface group, so that player-seasons with more observed points contribute proportionally to the year's surface estimate. For specialization, we compute a player-year "specialization index" by taking the variance across surfaces for a bundle of server-side metrics (currently `ace_rate`,

serve_only_rate, and avg_rally_shots) and then averaging those variances into a single scalar index; we summarize specialization over time by averaging the index across players within each year. In early results, the serve-only trends show the expected surface ordering (Grass highest, Clay lowest, Hard in between) and suggest an increase in serve-only rate on Hard courts into the modern era, consistent with rising serve dominance and shorter points on faster surfaces. The specialization index is sensitive to data coverage in early decades, but after applying coverage-based filters it appears relatively low and fairly stable from the 1990s onward with occasional spikes, which we interpret cautiously as a combination of true shifts and residual sampling variation.

5 Adjustments

We encountered three main obstacles and adjusted our workflow accordingly. First, point-by-point parsing at this scale was too slow and memory-intensive using a single pandas apply pattern, so we refactored to chunked processing and faster tuple-to-columns conversion to keep runtime manageable. Second, parquet output initially failed because the environment lacked a parquet engine (e.g., pyarrow or fastparquet), so we saved intermediate processed outputs as CSV to maintain compatibility, while keeping the option to switch back to parquet later. Third, early iterations of the aggregated dataset included many player-year-surface rows with all-zero rates, which largely reflected extremely small sample sizes (few charted points) and sparse early-era coverage; these rows created instability in yearly averages and exaggerated specialization spikes. To address this, we introduced minimum-points thresholds (e.g., requiring at least 300 points) for including player-year-surface rows in trend plots, and we restricted specialization calculations to player-years with at least two surfaces meeting a minimum-points-per-surface threshold. Going forward, we plan to report explicit coverage diagnostics (total charted points and unique players by year/surface) alongside trends, examine specialization distributions (median and upper quantiles) rather than only the mean, and, if necessary, restrict certain inferential analyses to eras with adequate coverage while still presenting descriptive plots over the full available timeframe.

6 Remaining Timeline

- Week 9 (by 3/6): Clean Graphs and data, Perform clustering analysis, Evaluate clustering robustness, Interpret cluster characteristics
- Week 10 (by 3/13): Analyze cluster evolution over time, Finalize statistical outputs, Begin writing report
- Finals Week (by 3/17): Complete full 8-page scientific report, Finalize GitHub repository (clean code + replication instructions), Prepare final presentation slides